

COMPUTATIONAL PLASTICITY

C.B. HOVEY

ABSTRACT. The concepts, formulation, and implementation for computational plasticity in one- and three-dimensions are explicated.

1. ONE-DIMENSIONAL PLASTICITY

We begin with plasticity in a single dimension to focus on concepts while being as mathematically simple as possible. After the concepts are clear, the extension of the mathematics from one- to three-dimensions can be readily made.

1.1. Concepts.

1.2. Formulation of Perfect Plasticity.

1.2.1. *Deformation Measures.* Let the body Ω has strain measure $\varepsilon : \Omega \rightarrow \mathbb{R}$ in one dimensions be defined as a ratio of change in length, that is the difference between the new (deformed, current configuration) length ℓ of body Ω and the original (undeformed, reference configuration) length L , divided by the original length,

$$\varepsilon \triangleq \frac{\ell - L}{L}. \quad (1)$$

The total strain ε in the body be defined as an additive decomposition of elastic strain ε^e and plastic strain ε^p ,

$$\varepsilon \triangleq \varepsilon^e + \varepsilon^p. \quad (2)$$

1.2.2. *Constitutive Law.* Then, the traditional stress-strain relationship, $\sigma = E\varepsilon^e$, can be rewritten as

$$\sigma = E(\varepsilon - \varepsilon^p). \quad (3)$$

A change in the configuration of the plastic element (figure to come) can only occur if the notion of **slip rate** is allowed. That is, *assume* that $\varepsilon^p(t)$ is a function of time on interval $[0, T] \subset \mathbb{R}$.

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